

Mobile ALOHA

Network Architecture

Deep Dive Analysis

ACT Policy: Action Chunking with Transformers

Comprehensive Analysis of Input Embedding, Encoder-Decoder Architecture,
and Action Decoding Mechanisms

**Based on: "Mobile ALOHA: Learning Bimanual Mobile Manipulation with
Low-Cost Whole-Body Teleoperation" (ICRA 2024)**

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1. Introduction & Overview

Mobile ALOHA ■ Berkeley ■ Stanford ■ ■■ ■■■ ■■ ■■■ ■■■■■ ■■■■■, ■■■ ■■■■ (\$32k) ■
■■■ ■■ ■■ ■■■ ■■■■ **Action Chunking with Transformers (ACT)** ■■■■■ ■■■■ ■■■.

Key Components:

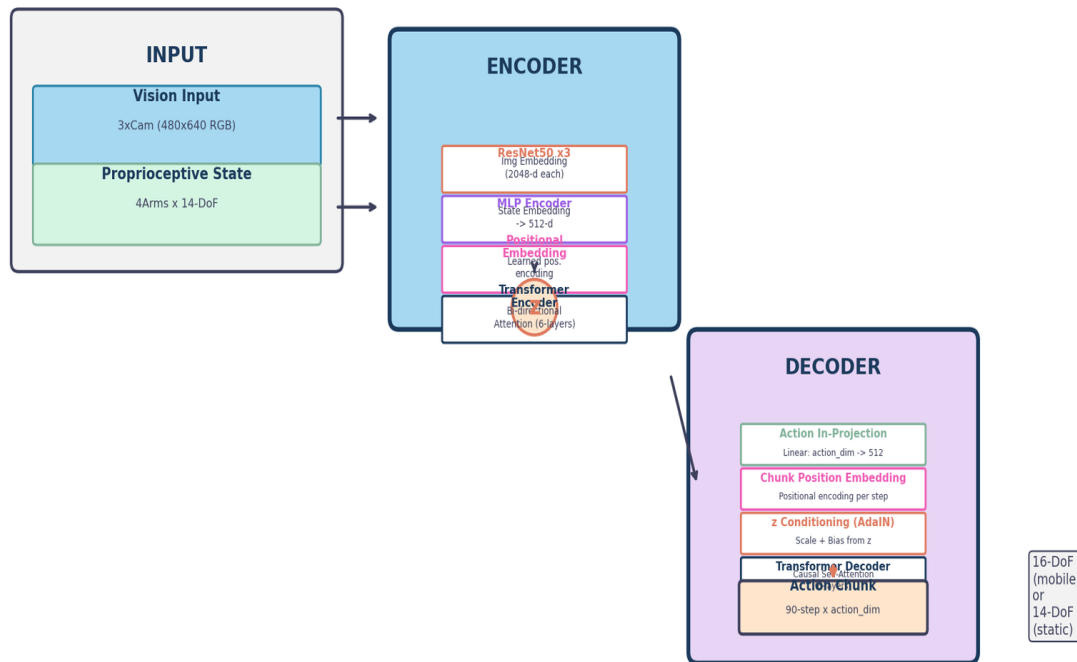
- * **Teleoperation System:** \$32k budget, 4-arm + mobile base, whole-body control
- * **ACT Policy:** CVAE-based transformer policy with action chunking
- * **Cross-embodiment Training:** Static ALOHA + Mobile ALOHA co-training
- * **High-frequency Control:** 50Hz output with 90-step action chunks (1.8s horizon)

2. Network Architecture

ACT  Conditional VAE (CVAE) . Encoder  observation  latent  z , Decoder  z  noisy action   action chunk .

Complete Architecture Diagram:

Mobile ALOHA ACT Policy: Complete Architecture



Action Space Detail

* Left Arm:

-> 6-joint + 1-gripper = 7-DoF

* Right Arm:

-> 6-joint + 1-gripper = 7-DoF

* Mobile Base:

-> v (linear) + w (angular) = 2-DoF

Model Size	Encoder Layers	Decoder Layers	Chunk Size	Training
~55M params	6 (bidirectional)	6 (causal)	90 steps (1.8s @ 50Hz)	CVAE recon + KL div

Architecture Flow:

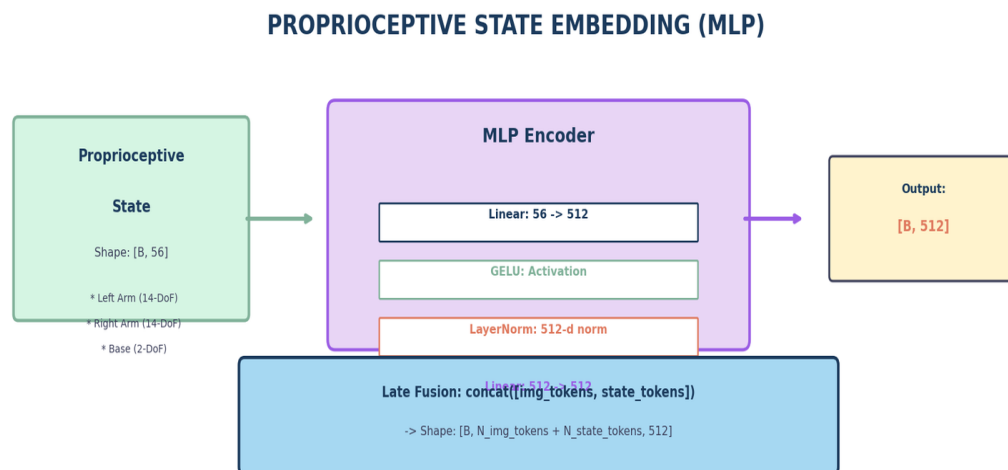
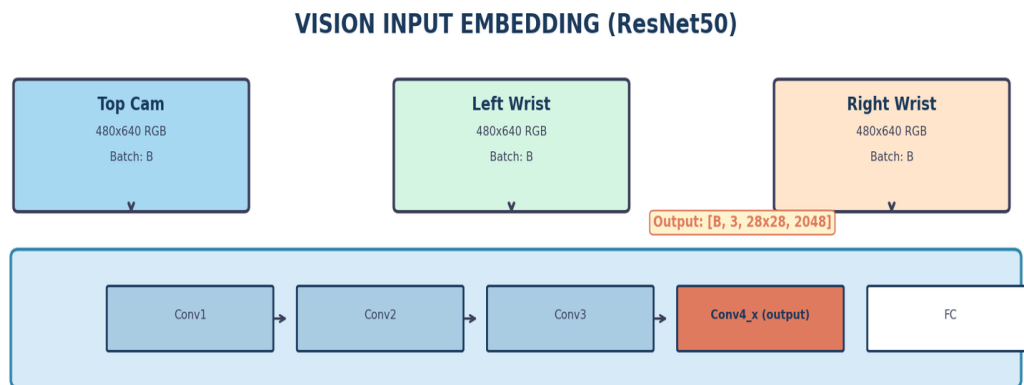
1. Observation (images + proprioceptive state) -> Encoder -> Latent z
2. z + Noisy Action Chunk -> Decoder -> Action Chunk [B, 90, action_dim]
3. Action chunk [step_start, step_end] -> [timestep_start, timestep_end] (Receding Horizon)

3. Input Embedding Details

3.1 Vision Input Pipeline

- * **Camera Configuration:** 3 cameras (Top + Left Wrist + Right Wrist)
- * **Resolution:** 480x640 RGB at 50Hz
- * **Encoder:** ResNet50 (pre-trained on ImageNet)
- * **Output per camera:** conv4_x feature -> [B, 28x28, 2048]
- * **Final vision tokens:** [B, 3x28x28, 2048] -> projected to [B, N_img, 512]

Vision Embedding Diagram:



Encoder Components:

- ## Latent Variable z :

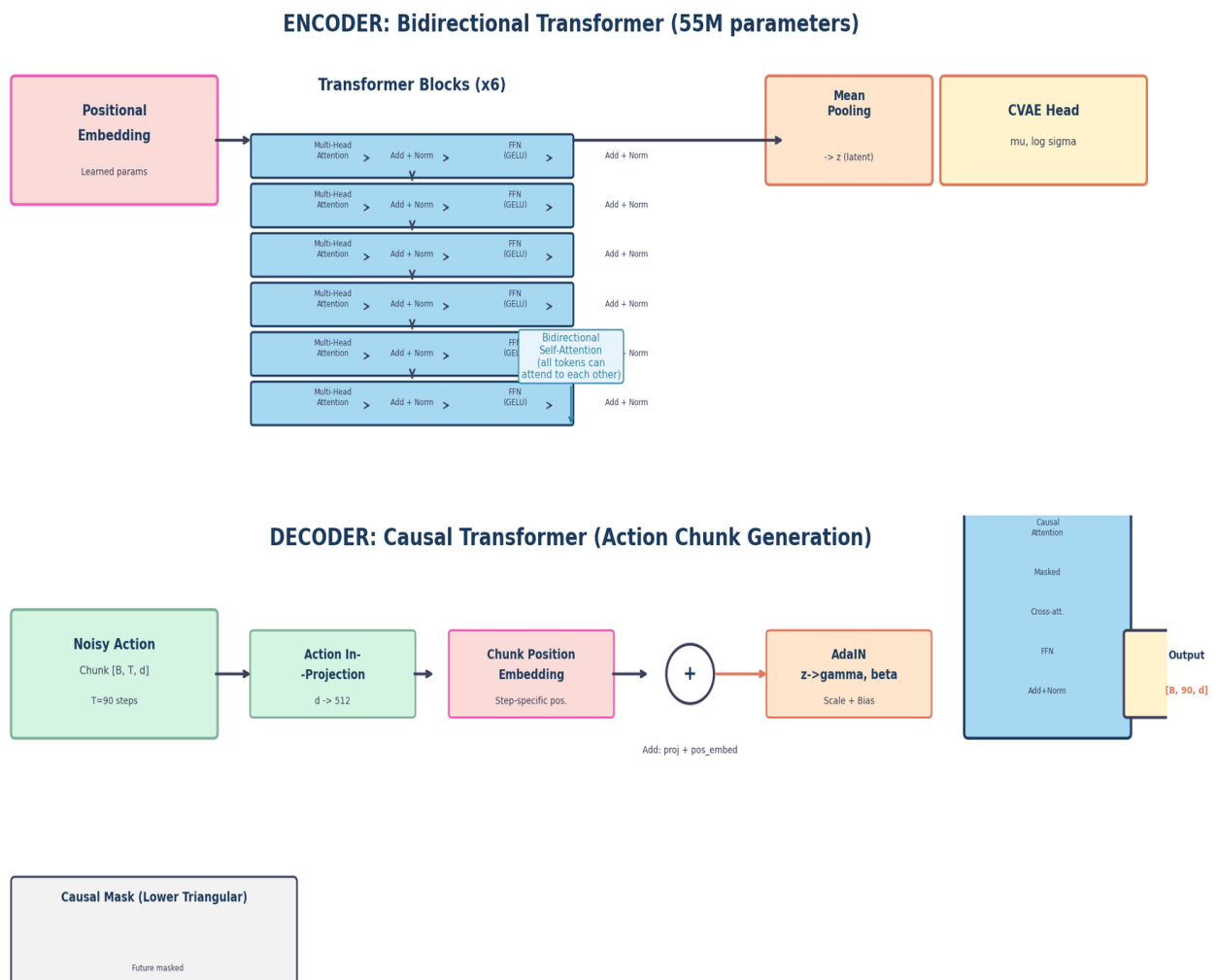
Inference ■: $z = 0$ (prior mean ■■)

5. Decoder Architecture

Decoder Structure:

- * **Action Input Projection:** Linear(action_dim -> 512)
- * **Chunk Position Embedding:** Step-specific positional encoding (90 steps)
- * **z Conditioning (AdaIN):** $z \rightarrow [\gamma, \beta] \rightarrow \text{action_tokens} * \gamma + \beta$
- * **Transformer Decoder:** 6 layers, causal (masked) self-attention
- * **Action Output:** Linear(512 -> action_dim) + LayerNorm

Decoder Architecture Diagram:



6. Training & Inference

Training Loss:

$$L_{\text{total}} = L_{\text{recon}} + \lambda \cdot L_{\text{kl}}$$

$L_{\text{recon}} = \text{MSE}(\text{action_pred}, \text{action_true})$ (reconstruction loss)

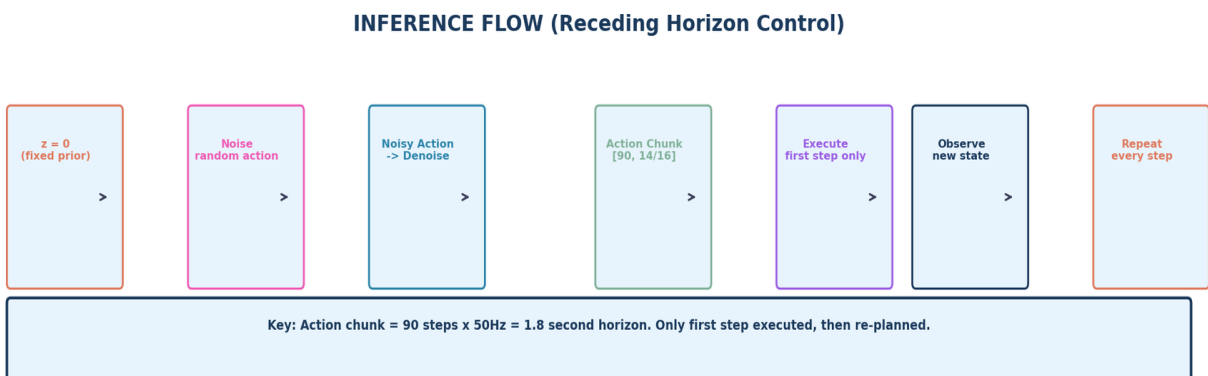
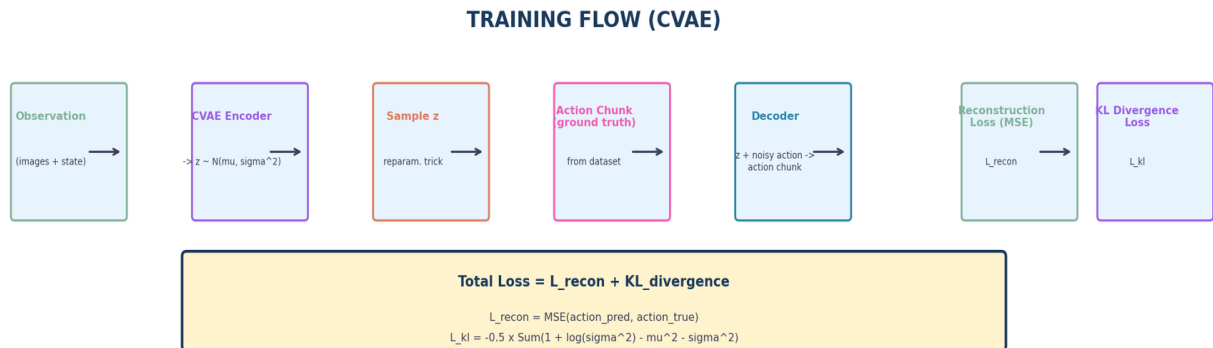
$L_{\text{kl}} = -0.5 \cdot \sum (1 + \log(\sigma^2) - \mu^2 - \sigma^2)$ (KL divergence to $N(0, I)$)

$\lambda = 1e-3$ (typically)

Inference Protocol (Receding Horizon):

- * 1. $z = 0$ (fixed prior, no encoder used)
- * 2. Generate noisy action: $a_{\text{noise}} \sim N(0, I)$
- * 3. Decoder $\rightarrow$ action_chunk = $[a_t, \dots, a_{t+89}]$
- * 4. Execute first step: robot.execute(a_t)
- * 5. Observe new state: $s_{t+1} = \text{robot.observe}()$
- * 6. Repeat from step 2 (full re-planning)

Training + Inference Flow:



7. Action Space Specification

Action space $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$:

Static ALOHA (14-DoF per arm):

- * Joint angles ($\theta_1, \dots, \theta_6$) - 6 DoF
- * Wrist angles ($\phi_1, \dots, \phi_4$) - 4 DoF
- * Gripper width (L, R) - 2 DoF

Mobile ALOHA (16-DoF total):

- * Left Arm: 6-joint vel + 1-gripper vel = 7-DoF
- * Right Arm: 6-joint vel + 1-gripper vel = 7-DoF
- * Mobile Base: v (linear) + ω (angular) = 2-DoF

8. Model Statistics & Summary

Category	Detail
Model Size	~55M parameters
Encoder	6-lay bidirectional Transformer
Decoder	6-lay causal Transformer
Chunk Size	90 steps (1.8 sec at 50Hz)
Control Frequency	50 Hz
Vision Encoder	ResNet50 x3 (pre-trained)
Latent Dimension	512
Training Loss	MSE reconstruction + KL divergence
Inference	Receding horizon (full re-plan each step)

Mobile ALOHA ACT CVAE Transformer observation latent space, action chunk. 90-step chunk (1.8) horizon, timestep receding horizon control.